CogLean: Towards a Cumulative, Machine-Checked Theory of Cognition

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Draft, October 2026

DRAFT FOR COMMENT — NOT FOR CIRCULATION[1]

Abstract

Cognitive science has many computational theories, but little cumulative theoretical knowledge. Theories are implemented as separate programs, so combining two of them usually means writing a third, and what a theory predicts is usually established by running it. We describe CogLean, a library of cognitive theory in the Lean proof assistant, organized on the model of Mathlib, the community library of formalized mathematics. CogLean is not a new theory of cognition. Its shared library, CogLib, holds representations, mechanisms, theory-neutral statements of phenomena, and theorems about them. Existing theories are expressed as choices of those mechanisms, and their consequences are proved, not simulated.

The central demonstration composes two independently formalized theories: EPAM's discrimination-net recognition (Feigenbaum and Simon) and the distributions-of-associations model of strategy choice (Siegler and Shrager). They are joined with a problem solver in the GPS and repair-theory tradition to produce a model of three-digit addition. The components, and most of the theorems the composition relies on, were established before it; the composition adds linking lemmas but no new psychological assumptions. The composed model computes all of its answers; no route is given the correct sum. One consequence is the *confident error*: a problem confused with a familiar one gets the familiar problem's answer, wrongly, with exactly as much use of the backup procedure as the familiar problem itself, which is rare when retrieval on the familiar problem usually succeeds. Neither theory makes this prediction alone. Others are that improving the fallback solver cannot reduce such errors, and that a confused problem has two sources of error. Replacing EPAM's memory with ACT-R's declarative memory, and changing nothing else, removes the conditions under which the confident-error theorem applies; ACT-R still produces graded errors of a similar kind, and laws of memory that one memory has and the other lacks explain the difference. Formalization alone also produced a correction to one of the original theories: making a distribution of associations more peaked can *increase* overt strategy use, contrary to an informal claim of the original chapter.

We separate results that hold for every setting of the mechanisms from results that need stated conditions, and both from exact values at illustrative settings. We also report what an independent, blind simulation found that the proofs did not: theorem hypotheses that were never realized in the tested learning conditions, and an unstated assumption on which a predicted learning sequence depends.

[1]This draft has *not* been carefully checked by the author. It is likely to contain erroneous citations and details. Please do not cite or distribute it.

1 Introduction

Cognitive science has accumulated many computational theories of memory, learning, problem solving and skill. It has accumulated much less theory that can be *reused*. A theory of memory developed for verbal learning usually has to be reimplemented before it can take part in a model of arithmetic. When two theories are combined, the result is a new program, whose behavior must be explored afresh by simulation. Newell's complaint that the field asks nature twenty questions without assembling the answers [1], and his call for unified theories [2], are about this problem. So, more recently, are arguments that computational modeling forces theories to be made explicit [3] and that psychology needs explanatory theories, not only tests [4].

Mathematics faced a version of the same problem and has made progress on it. Mathlib [5], the community library of formalized mathematics for the Lean proof assistant [6], does not propose new mathematics. It provides one shared formal environment, in which definitions are made once, every result is machine-checked, and later work builds on earlier results by citing them. A theorem about groups, once proved, is available to everyone who uses groups.

This paper asks whether the same organization can work for cognitive theory:

> *If independently developed cognitive theories are represented in a common formal library, can we derive scientifically meaningful consequences from their composition, consequences that would otherwise require building and investigating a new integrated model from scratch?*

We describe CogLean [7], an initial library built to test this, and give a concrete, checked answer in one case. The case is three-digit addition, composed from EPAM's discrimination nets [8, 9], Siegler and Shrager's distributions-of-associations (DOA) model of strategy choice [10], and a problem solver in the tradition of GPS and repair theory [11, 12]. Arithmetic is the demonstration, not the objective. We are not proposing a new theory of children's arithmetic. Multi-digit addition is a familiar task, complex enough to need several mechanisms, and two of the needed theories already exist.

The argument is organized around three capabilities, in increasing order of importance.

1. **Formalization.** Existing theories are stated as definitions, postulates and theorems. Each theorem's assumptions are explicit.

2. **Composition.** Formalized mechanisms are reused in new contexts, and so are the *theorems* about them. A library of reusable cognitive algorithms would be useful software. A library of reusable *results* is a contribution to theory.

3. **Discovery.** Composition establishes consequences that the original theories' authors did not state. Formalization also exposes unstated assumptions, results that depend on parameters, results that hold for whole classes of models, and conditions under which theories diverge. It can also correct a theory: in Section 3.3, an informal claim of the DOA chapter turns out not to follow from the model's own postulates.

Lean checks that conclusions follow from premises, not that the premises describe people or that a valid derivation explains anything (Section 7).

2 CogLean

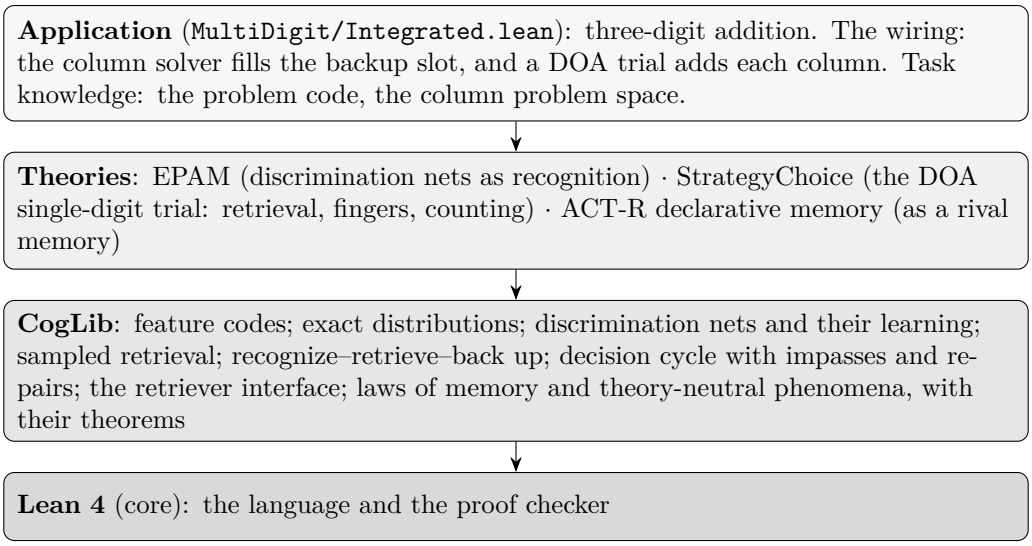


Figure 1: The layers of CogLean. Arrows point from a layer to what it imports. Each layer's theorems are available to the layers above it.

2.1 Organization

CogLean has four layers (Figure 1).

- **Lean**, the formal language and proof checker. CogLean uses only Lean's core, not Mathlib, and every probability is an exact rational number.
- **CogLib**, the general library. It holds:
  - representations: feature codes, finite probability distributions, semantic networks;
  - memory structures: discrimination nets, an ACT-R-style activation memory;
  - processes: discrimination learning, sampled retrieval with a confidence criterion, "recognize, then retrieve, else back up", and a decision cycle with impasses and repairs;
  - an interface any memory can implement (a *retriever*: study an item, retrieve a distribution over stored items);
  - theory-neutral statements of phenomena and laws, such as the serial-position effect, or the law that retrieval of a studied item is certain.

  CogLib imports no theory.
- **Theories**, each a package that states its postulates as choices of CogLib constructs, cited to their sources, and proves its phenomena.
- **Applications**, models of particular tasks assembled from theories and library mechanisms, with the task's own knowledge.

The contribution is not a programming language or a new syntax for cognitive models. Any model could be written in Lean as a program. What CogLean contributes is an *organization* of theoretical knowledge: mechanisms defined once, phenomena stated independently of any mechanism, and theorems relating them, all in one place where later work can cite them. CogLean is far from Mathlib's scale or maturity. It is a demonstration that the approach can work, not a finished library.

2.2 Mechanisms, phenomena, theorems and conditions

Four kinds of object must be kept apart.

1. **Mechanisms and their postulates**: definitions, such as how a discrimination net sorts a code, or how retrieval samples an answer in proportion to its associative strength.

2. **Phenomena**: predicates, stated without reference to any mechanism. A predicate asserts nothing. It is a target, which some mechanism may or may not be proved to satisfy.

3. **Theorems**: proofs that a mechanism produces behavior satisfying a predicate.

4. **Conditions**: the hypotheses under which a theorem holds. Some are situational ("problem x is confused with problem q"), some are empirical premises that remain unexplained ("q's associations are sharp"), and some are well-formedness requirements ("the strengths have positive total").

A theorem that has a phenomenon among its hypotheses is a relation between phenomena, not a prediction of a model. It becomes a prediction only when some mechanism discharges the hypothesis by proving it. CogLean's catalog of theorems, generated from Lean, marks such conditional results mechanically [7].

Shared definitions make theories comparable at the level of proofs. For example, CogLib proves once that any learner whose learning never disturbs what it already knows reaches a learning criterion (`criterion_of_nonInterference`). It separately proves that discrimination-net learning never disturbs prior learning (`netLearner_nonInterference`). EPAM's learning to criterion then follows in a line, and so will that of any later theory whose learning is non-interfering.

3 Two theories, formalized independently

3.1 EPAM: recognition by discrimination

EPAM [8, 9] recognizes a stimulus by sorting its code, a vector of features, down a discrimination net of tests to a stored image. When a stimulus is confused with a stored one, learning grows a test at that point, on the first feature in a noticing order where the two differ. We formalize the propositions of Feigenbaum's report, not the original IPL-V program. CogLib proves, for every net, code and noticing order:

- recognition depends only on the features tested on the path a code follows;
- learning a stimulus never changes the recognition of another (`learn_preserves`);
- a confusion agrees with the stored image on every feature tested on its path (`confusion_agrees_on_path`);
- every net learned from an empty net stores each image once and recognizes each as itself (`learnAll_clean`).

3.2 DOA: distributions of associations

In the DOA model [10], a child adding single digits has associations of varying strength between each problem and possible answers.

1. On each trial the child has a confidence criterion c and a search length n.
2. A search samples an answer in proportion to strength, and states it if its strength exceeds c, for up to n searches.
3. If retrieval is exhausted, the child puts up fingers, which adds strength to the answer that many fingers represents, and retrieves once more.
4. Failing that, the child counts the fingers, with fixed chances of skipping or double-counting an object.
5. Every stated answer is strengthened, a correct one more than an incorrect one.

Let R be the chance that one search clears the criterion. StrategyChoice proves for every distribution, criterion and search length:

- retrieval is used with probability $1 - (1 - R)^n$ (`SC.retrieval_prob`);
- the error rate among retrieval trials is the wrong answers' share of the strength above the criterion, whatever the search length (`SC.retrieval_error_share`);
- neither depends on the backup strategies (`SC.retrieval_independent_of_backup`);
- the error rate among counting trials does not depend on the associations at all, which is the chapter's "nonintuitive prediction" (`SC.counting_error_share`).

3.3 A discovery from formalization alone

The DOA chapter's central idea is that the peakedness of a problem's distribution of associations governs both accuracy and strategy use. It states that "a peaked distribution results in less frequent use of overt strategies" [10, p. 245]. Formalizing the model showed that half of this follows from the postulates and half does not.

- **What follows.** Moving strength to the correct answer never raises the error rate on retrieval trials, at any criterion or search length (`SC.peaking_lowers_retrieval_errors`), and neither does learning from a correct answer (`SC.correct_strengthening_lowers_errors`). These are general theorems.
- **What does not.** Peaking can *raise* overt strategy use. Take $1 + 1$ with answers 2, 3 and 1, and strengths $.38, .31, .31$. Move one hundredth of strength from the wrong answer 3 to the correct answer 2, which is already the strongest: $.39, .30, .31$. With search length 1, averaged over the chapter's own twelve confidence criteria, predicted overt strategy use rises from 7/12 to 73/120 (`SC.peaking_can_raise_overt_use`). The reason is that the weakened wrong answer falls to the criterion .30, so it no longer clears it. Searches that used to state it now fail, and the child turns to fingers.

The chapter's claim is not wrong in general; it needs a condition the chapter does not state, roughly that no competitor sits near a criterion. Nothing about this required composition or new mechanisms. It came from stating the postulates precisely and asking whether the informal claim followed. It is the clearest case in the library of formalization correcting a theory's description of its own consequences.

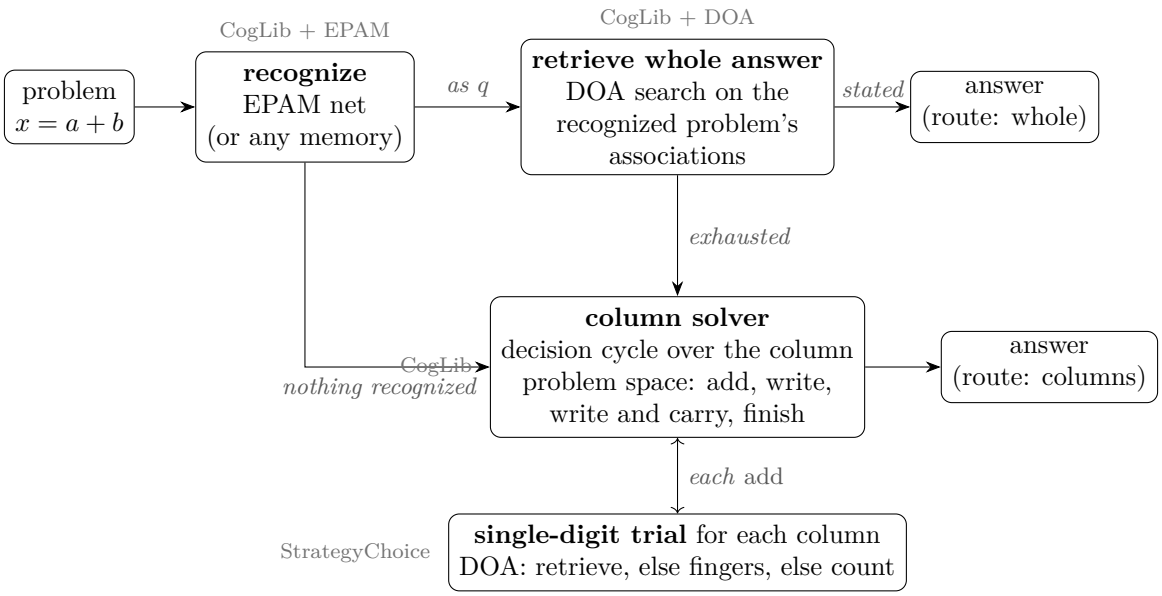


Figure 2: One trial of the integrated model (`MD.Integrated.itrial`). Grey labels give the source of each mechanism. When x is recognized as a familiar q, the whole answer is retrieved from q's associations. The solver runs only when recognition fails or retrieval is exhausted, and it computes the answer from single-digit trials and a carry rule.

4 Composing theories: multi-digit addition

4.1 The composition

A child adding two three-digit numbers may know the answer to a familiar problem such as 100+100, or may work it out column by column. The model nests DOA's architecture one level up (Figure 2). It has six parts.

1. **A problem code**: features for each digit of each addend. Agreeing on a digit's features means having the same digit (postulate M1, `MultiDigit/Postulates.lean`).
2. **Recognition** by an EPAM net that stores familiar problems (M2).
3. **Whole-problem retrieval**: DOA sampling from the recognized problem's associations (M3).
4. **A column solver**, invoked when nothing is recognized or retrieval is exhausted (M4). It is a problem space with operators to add a column, write a digit, write and carry, and finish, run by a procedure (`MultiDigit/Solver.lean`).
5. **A DOA single-digit trial** for each column's addition, with retrieval, fingers and counting (`MD.doaFact`, wrapping `SC.trial`).
6. **CogLib's decision cycle**, which sequences the operators and handles carrying, impasses and repairs (`CogLib/Process/ProblemSolving.lean`).

In Lean, the composition is three definitions (`MultiDigit/Integrated.lean`):

```
def solverBackup (F : Facts) (p : Problem) : Dist Nat :=
  Solver.answers F.fact Solver.full Solver.lenient p
def itrial E N str as cc n F x := trial E N str as cc n (solverBackup F) x
```

Here `F.fact` is the single-digit DOA trial, `Solver.full` the full column procedure, and `trial` the recognize-and-retrieve trial from `MultiDigit/Postulates.lean`. The integrated model introduces no new mechanism. Its only correct arithmetic knowledge is in the single-digit associations, which are peaked on each digit pair's sum, as in DOA, and in the carry rule.

4.2 Why the integrated model is stronger evidence

An earlier version of the example, still in the repository (`MultiDigit/Example.lean`), checked the general theorems with a backup that simply returned the correct sum, `Dist.pure p.sum`. That suffices to instantiate theorems that hold for *every* backup. But it supplies the answer instead of computing it, so it is not evidence that the mechanisms compose. In the integrated model, nothing is handed the answer. The answer 250 to $150+100$ is assembled by the solver from three DOA trials and a carry rule, and it is never certain. So the composition must work at the level of mechanisms (the parts fit and run together) and at the level of theorems (the parts' results transfer to the whole).

4.3 The executable backup: arithmetic as general problem solving

When whole-problem retrieval fails, the integrated model invokes an executable column-addition solver. It is an instance of CogLib's general problem-solving framework (`CogLib/Process/ProblemSolving.lean`), not a separate arithmetic program. The framework is task-general:

- a *problem space* has states and operators; each operator has a precondition and an effect, and an effect is a distribution over states, so an operator can include a probabilistic process;
- a *decision cycle* runs a procedure that proposes an operator in each state; a proposed operator whose precondition is violated is an *impasse*, which repair heuristics, filtered by critics, resolve [12].

The arithmetic instance (`MultiDigit/Solver.lean`) defines operators to add a column, write its digit, write and carry, and finish. The procedure `full` proposes them as the calculation progresses, units first. `Solver.answers` runs CogLib's decision cycle, for a bounded number of cycles (eight, enough for three columns), and reads off the written answer. In `solverBackup`, `F.fact` supplies the single-digit process and `Solver.lenient` the repair policy; with the full procedure, no impasse arises.

Each *add* operator performs a DOA trial on its column's digits and returns a distribution of possible digit sums. The framework composes such steps with `Dist.bind`, which carries every possible outcome, with its probability, through the operations that follow. Nothing is sampled, and nothing consults the correct answer. The result is a distribution over complete multi-digit answers, errors included.

CogLib also implements GPS-style means–ends analysis, with goals, subgoals and backtracking. It has been instantiated for the same arithmetic problem space (`MultiDigit/SolverGPS.lean`), where it is proved to work units first and to answer every problem exactly when the single-digit facts are exact. The integrated demonstration uses the decision cycle, not means–ends analysis. The GPS results are complementary: they concern goal-directed operator selection.

4.4 Composition of theorems

Three kinds of reuse make the composition deductive rather than merely executable.

A bridge. There are two executable descriptions of column addition. `columns` (`MultiDigit/Columns.lean`) is a direct implementation of probabilistic column arithmetic. `answers` performs the same task on the decision cycle just described. The bridge theorem proves that they produce identical answer distributions, for every problem and every single-digit process:

```
theorem bridge (fact : Nat → Nat → Dist Nat) (p : Problem) :
    answers fact full lenient p = columns fact p
```

The equivalence is proved, not established by testing examples. Every theorem proved about `columns` therefore holds for the solver; for instance, that a column error is a single-digit error moved to its place when the other columns are exact (`MD.column_error_is_doa_error`). The bridge equates output distributions, not internal processes. As accounts of what a child does, the two descriptions still differ.

Associativity. Each arrow in Figure 2 is a two-stage draw of finite distributions. CogLib proves that such draws regroup (`Dist.bind_assoc`), so a theorem about a component holds where the component sits inside the whole.

Well-formedness, inherited. The general confident-errors theorem needs the backup to have total probability 1. For the integrated model this is proved, not assumed, by chaining results from three packages (`mass_solverBackup`):

- DOA retrieval is proper (CogLib, `mass_search`);
- a whole DOA trial is proper (StrategyChoice, `SC.mass_trial`);
- the column procedure is proper when its facts are (MultiDigit, `mass_columns`);
- the solver equals the column procedure (`Solver.bridge`).

A parallel chain proves that every weight is nonnegative, given nonnegative strengths and rates (`nonneg_search`, `SC.nonneg_trial`, `nonneg_solverBackup`, `nonneg_itrial`). Both are needed: CogLib's distributions are lists of rational weights, and total mass 1 alone does not exclude negative weights. These lemmas contribute no psychology. They are what allows theorems written for an abstract backup to apply to a concrete, computing one, and the integrated trial to be read as a probability distribution.

5 Derived consequences

We present four results, chosen because their explanatory status differs. The first is explained in detail.

5.1 Confident errors

In DOA alone, a problem's errors and its use of backup strategies come from the same distribution. A flat distribution of associations produces both more retrieval errors and more backup use, so errors and hesitation go together. Recognition breaks this tie. A problem confused with a familiar one is retrieved from the familiar problem's distribution: with that problem's low backup use, and with all of its answers, which are wrong for the confused problem.

The theorem, as stated for the integrated model (lightly abridged):

```
theorem integrated_confident_errors (E : Encoding) (N : Net) (str ...) (F : Facts)
    (hF : F.Proper) (x q : Problem) (hT : ∀ y, total as (str y) ≠ 0)
    (hq : N.select (E.code q) = some (E.code q))      -- q is recognized as itself
    (hx : N.select (E.code x) = some (E.code q))      -- x is confused with q
    (hs : Sharp str cc (E.code q) q)                  -- only q's sum clears the criterion
    (hne : x.sum ≠ q.sum) :
    ConfusionErrorPattern (itrial ...) wholeWrong isWhole isColumns x q

-- CogLib's statement of the phenomenon, for any trial and outcomes:
def ConfusionErrorPattern trial wholeWrong isWhole isBackup x q : Prop :=
  prob (trial q) (wholeWrong q) = 0 ∧                              -- (i)
  prob (trial x) (wholeWrong x) = prob (trial x) isWhole ∧         -- (ii)
  prob (trial x) isBackup = prob (trial q) isBackup                -- (iii)
```

The phenomenon is defined once, in CogLib (`Phenomena/ConfusionErrors`), in terms of observable outcomes only. It names no mechanism, and none of the theorem's conditions (confusion, sharpness, proper distributions) is part of it; they are hypotheses of the theorems that establish it. In words: (i) whole-problem retrieval never errs on q; (ii) every whole-problem answer to x is wrong; (iii) x goes to the column solver exactly as often as q does. The confused problem is answered wrongly with the same fluency as the familiar one.

Equal backup use is not low backup use. Whether the errors are *confident*, in the sense that the solver is rarely invoked, is a separate, quantitative matter. The rate is $(1 - R_q)^n$, where R_q is the chance that one search of q's associations clears the criterion (`integrated_backup_rate`). If $R_q \geq r$, the confused problem reaches the solver with probability at most $(1 - r)^n$ (`integrated_backup_le`). In the illustrative setting, $R_q = 45/46$ and $n = 1$, so the solver is reached $1/46$ of the time. With flat associations on q, the same theorem would hold, and the errors would not be confident. CogLib accordingly states a second, stronger phenomenon, `ConfidentErrors r`: the pattern with backup use at most r, where the bound is an argument, not a fixed threshold. The integrated model has it with $r = (1 - r_q)^n$ for any lower bound r_q on R_q (`integrated_confident_errors_le`), and the setting has it with $r = 1/46$ (`setting_confident_errors_bound`).

What is assumed. The postulates are the mechanisms: the net, DOA retrieval and the solver. The conditions are of three kinds.

- *Recognition conditions* (`hq`, `hx`): q is recognized as itself and x is confused with it. These are properties of a particular net. They are not arbitrary: every net EPAM learns recognizes its images as themselves (`learnAll_clean`), and the net in our setting, learned by EPAM from $100+100$ and $100+200$, is verified to confuse $150+100$ with $100+100$ (`Example.net_facts`).
- *Sharpness* (`hs`): only q's correct sum clears the criterion. This is an empirical premise that the library does not yet derive from learning.

- *Well-formedness* (`hF`, `hT`): strengths have positive totals.

No hypothesis mentions confident errors, or any relation between errors and backup use.

What is reused. The proof is one line. It applies `MD.confident_errors`, the theorem for an arbitrary backup, with `mass_solverBackup` for its well-formedness hypotheses. `MD.confident_errors` is in turn a relabelling of CogLib's `RR.confident_errors`, which holds for any task with problems, a familiar set and a backup. That theorem combines three earlier lemmas:

- a problem recognized as a sharp q gets only q's answer (`confusion_gives_neighbours_answer`);
- so all its whole-problem answers are wrong when the sums differ (`wrong_eq_whole_of_sharp`);
- backup use after certain recognition as y is $(1 - R_y)^n$, which depends only on y (`backup_prob_certain`).

The last lemma is DOA's retrieval-probability result, lifted to a recognized item.

Generalizing over memories. The same pattern is proved from laws of memory instead of from a particular net (`RR.confident_errors_of_laws`). Any memory with three properties produces confident errors whenever a stranger can be taken for a studied item with sharp associations:

- *determinacy*: retrieval calls up a single item for certain;
- *certainty*: a studied item retrieves itself;
- *stability*: studying one item does not change another's retrieval of itself.

This is a result about a class of models, not about one model.

5.2 Whole-problem errors do not depend on the solver

Once x is recognized as q, the chance of a whole-problem answer with any property is that of retrieval on q's associations, whatever the backup (`whole_independent_of_solver`). This is a general consequence of the composition, at every setting. Its psychological content is a prediction about intervention: improving column skill, for instance by teaching carrying, cannot reduce errors that arise from mistaken recognition. Only learning that changes recognition or associations can.

5.3 Two sources of error

Under the same conditions as confident errors, the integrated model distinguishes two sources of error on a confused problem (`errors_two_sources`).

- **Whole-route errors** are always the confused neighbour's answer (here 200), whenever that route is taken.
- **Column-route errors** occur with probability equal to the chance that retrieval is exhausted times the solver's own error rate on x, which comes entirely from the single-digit DOA trials.

This result is *compositional*. Neither recognition nor DOA says it alone, and it would not hold with the old backup, which never erred.

The two sources tend to leave different signatures, but only the first is guaranteed. A whole-route error is always the confused neighbour's sum. A column-route error has a simple signature only when one column errs: if the other columns are exact, the problem is wrong exactly when that column's DOA trial is, and off by $10^j\delta$ when column j errs by δ (`solver_error_is_doa_error`). Several columns can err on the same trial, and a column error can propagate through a carry. Errors can even cancel. Exact single-digit facts in every column are sufficient for an exact answer (`solver_exact_of_exact`), but not necessary: for $150 + 100$, a units result of 10 and a tens result of 4 still give 250 (`compensating_errors`).

5.4 Memory substitution

Because recognition is stated for any retriever, EPAM's net can be replaced by ACT-R's declarative memory [13], with everything else unchanged: the encoding, the associations, the criterion, the search length and the solver. CogLib implements an ACT-R-style memory with base-level activation, partial matching and a retrieval threshold, for decay 1 and temperature 1, so that probabilities stay rational (`CogLib/Memory/Activation.lean`).

The two memories differ in the laws just listed. Both cells are proved, for every parameter setting:

- the discrimination net has determinacy, certainty and stability (`Matrix.Discrimination.determinate`, `.certainty`, `.stability`);
- ACT-R's memory has none of them (`Matrix.Activation.never_determinate`, `never_certainty`, `never_stability`).

So the general confident-errors theorem applies to EPAM and never to ACT-R: its hypotheses cannot be met, because nothing is retrieved for certain (`Act.retrieve_not_certain`). That the theorem does not apply does not show that ACT-R cannot produce similar behavior, and at the illustrative setting it does: $150 + 100$ is answered 200 by retrieval with probability .43. What fails is the exact pattern, and at the setting this is proved, not just observed: with ACT-R's memory the trial does not have `ConfusionErrorPattern` (`setting_actr_not_pattern`). Retrieval of the confused problem is graded, and its solver use differs from the familiar problem's (383/1058 against 107/782), so (iii) fails. This is a counterexample at one setting, not a proof that ACT-R's memory lacks the pattern at every setting. A graded analogue of confident errors for memories like ACT-R's would need its own theorem, and its own laws. Table 1 shows the consequences at one setting.

With EPAM, the confusion is certain and confident errors dominate. With ACT-R, the confusion is graded. $150 + 100$ can retrieve either familiar problem, or fail. It therefore reaches the solver far more often and is answered correctly about a quarter of the time. ACT-R also predicts interference: studying $100 + 300$ roughly doubles the retrieval errors on the practiced $100 + 100$ (`act_interference`), and raises its overall errors from about a third to about three fifths. EPAM predicts no interference, for every net and every backup (`MD.learning_preserves_trial`). The difference is between kinds of memory, and the laws say which property of memory produces it.

5.5 One memory for whole problems and column facts

In the integrated model, the solver's single-digit facts come from a DOA association table of their own, separate from the memory that recognizes whole problems. A child has one memory. `MultiDigit/SharedMemory.lean` removes the separation, again by composition alone. Each

	EPAM net	ACT-R memory
$150 + 100$ retrieves $100 + 100$	certainly	with probability $10/23$
$150 + 100$ answered 200 by whole-problem retrieval	$45/46$	$225/529 \approx .43$
$150 + 100$ answered 300 (taken for $100 + 200$)	0	$225/1058 \approx .21$
$150 + 100$ reaches the column solver	$1/46$	$383/1058 \approx .36$
$150 + 100$ answered correctly (250), by the solver	$\approx .014$	$\approx .238$
practiced $100+100$: errors by whole-problem retrieval	0	$225/782 \approx .29$
practiced $100 + 100$: errors overall, with the solver	$\approx .0073$	$\approx .334$
after also studying $100+300$: whole-problem retrieval	0	$3375/5842 \approx .58$
after also studying $100 + 300$: overall	$\approx .0073$ (unchanged)	$\approx .616$
for every setting (general theorems)	confident-error theorem applies; practice never disturbed	its hypotheses never hold; interference possible

Table 1: The same application with two memories, at one illustrative setting: the binary-coded encoding, both memories having studied $100 + 100$ and then $100 + 200$, the same associations, criterion $1/2$, search length 1, and the same solver. ACT-R's parameters are a mismatch penalty $\mu = 1/2$ over 24 features and a failure weight $\theta = 1/10$. All values are exact and proved in Lean. Rows about whole-problem retrieval do not depend on the backup (`whole_independent_of_solver`), so they are the same in `ExampleACTR.lean`, which uses an exact backup. Rows about answers overall use the integrated model, with its fallible solver (`setting_actr_x`, `setting_practiced_errors`). EPAM's overall errors on 100+100 are the solver's own; after studying 100+300 they are unchanged, by `MD.learning_preserves_trial`, which holds for every backup. The last row states what holds at every setting.

column's digit pair $d + e$ is coded in the problems' own code space, as the problem $d + e$. A column's addition is then recognize–retrieve–back up on that pair (`memFact`): the *same* retriever that recognizes whole problems calls up a stored code, and retrieval searches its associations in the same strength table. When nothing clears the criterion, the column falls back on DOA's fingers and counting. The memory becomes a single parameter, used at both levels; the solver itself is unchanged, since it is parametric in its facts.

Three results follow from theorems already in the library:

- **The separate table is a special case.** If the memory recognizes a digit pair as itself, the column fact *is* the DOA fact on that pair's associations (`memFact_self`). So when each column's pair is recognized as itself, the shared model is the integrated model (`strial_reduces`, `strial_net_reduces`), and every theorem of §5 holds of it.

- **Confident errors one level down.** A digit pair taken for another, whose associations are sharp, shows the confusion-error pattern: every retrieved answer is the other pair's sum, with no more fallback than the other pair (`column_confident_errors`). The proof applies `RR.confident_errors` to digit pairs; it needed no new lemma.

- **The two memories diverge further.** At the setting (the memory studies $0 + 0$, $5 + 0$ and $1 + 1$, then the two problems), EPAM's net recognizes all five as themselves, and every value

Kind	Results	What they rest on
Library facts: properties of mechanisms, needed for reuse	the solver equals the column procedure (the bridge); the solver and the integrated trial are probability distributions; merged evaluation is the same distribution	the definitions, and nonnegative strengths with positive totals
Compositional deductions: hold for every encoding, net, strength set and DOA setting	whole-problem errors are independent of the solver; the column route factors as exhaustion times the solver; the solver's rate of use is $(1 - R_q)^n$	the composed mechanisms alone
Conditional: hold given stated conditions	confident errors; two sources of error; a solver error as a single-digit error in place (other columns exact); exactness of the solver (exact facts: sufficient, not necessary); confident errors for any memory with three laws	recognition conditions (derivable for EPAM nets, verified for the learned net); sharp associations (an unexplained empirical premise); exact columns; for the last, the laws themselves
At chosen settings: exact values	150 + 100: 200 by retrieval 45/46; solver reached 1/46; solver correct ≈ .656; correct overall ≈ .0143; Table 1	an encoding, a learned net, associations, criterion and search length chosen for illustration

Table 2: The integrated model's results, by kind. Lean names are in Appendix A.

is the integrated model's (`setting_epam_values`). ACT-R's memory does not: digit pairs and whole problems compete for retrieval. $0+0$ calls up $5+0$ with probability 15/197 and is answered 5 with probability 120/2167; the solver answers $150+100$ correctly with probability about .385, against .656 with the separate table; and the practiced $100 + 100$ is answered wrongly about .386 of the time, against .334 (`setting_act_pairs`, `setting_act_values`).

These ACT-R values carry the limitation stated below: retrieval does not change ACT-R's memory here. And part of ACT-R's loss comes from the fallback to finger counting, which a child doing three-digit addition would not use.

6 Three kinds of results

Table 2 sorts the results by what they rest on. A reader should be able to tell which statements are library facts about mechanisms, which are deductions from the composition, which hold only given further conditions, and which are numbers at chosen settings. Not every theorem is a psychological prediction, and none here is an empirical finding: the conditional and compositional results become empirical predictions about people only once their conditions can be identified in people and the predictions tested.

The illustrative setting. The encoding codes each digit of each addend in four bits. The net is learned by EPAM from $100 + 100$ and $100 + 200$, scanning hundreds digits first, so it tests only the bit where those two problems differ. The whole-problem associations give strength .90 to each familiar problem's correct sum and .01 to other answers, with criterion .50 and search length 1. Every digit pair has the same single-digit associations: 4/5 on its sum, 3/25 on the next number

in the counting sequence, and 1/100 on each other answer from 0 to 19. The single-digit criterion is 1/10, the search length 1, and finger elaboration, skipping and double-counting are each 1/20.

At this setting, 150+100 is answered 200 by whole-problem retrieval with probability 45/46 and reaches the solver with probability 1/46. The solver alone answers 250 with probability $\approx .656$. Its main errors are 260, 350 and 251, each $\approx .1$: a tens error, a hundreds error (the counting-sequence neighbour of $1 + 1$) and a units error. So $150 + 100$ is answered correctly with probability $\approx .0143$ (`setting_routes`, `setting_solver_x`, `setting_overall_x`). The familiar $100 + 100$ is answered correctly with probability $\approx .993$.

These numbers are exact consequences of the setting, and they show that the conditional theorems' hypotheses can be met by a learned net. They are *not* fitted to children's data and are not predictions about children. (Computing them needed a merged evaluator, proved to be the same distribution as the model, `SC.same_trialMerged`: written out, counting k objects lists 3^k outcomes.)

7 The scientific status of proofs

Validity is not empirical adequacy. A proof establishes that a conclusion follows from its premises. It does not establish that the premises are true of people. Confident errors follow from CogLean's formalization of EPAM and DOA. Whether children make them, and whether they make them for this reason, are empirical questions.

Validity is not explanation either. For a theorem to be an explanation, its premises must characterize mechanisms independently, and not restate the phenomenon. An earlier version of CogLib's serial-position result illustrates the difference. Its mechanism was a schedule on which item i was learned on trial twice its distance from the nearer end. Proving that this produces a bowed error curve was valid, but the bow was in the definition. The current version derives the order of learning from a rule over any set of anchors: items nearer an anchor are attended first. The same rule then has a second consequence, the isolation (von Restorff) effect, when a distinctive item serves as an anchor (`CogLib/Control/Serial.lean`). The result is still close to its premises. But one mechanism now answers to two phenomena, which is a real test.

The repository records an explicit audit of this kind for its headline results (`notes/EXPLANATORY_AUDIT.md`). It sorts each result by how far it lies from its premises:

- *definitional*: a restatement of the definitions;
- *close*: the pattern is largely in the mechanism;
- *derived*: it follows from premises that do not mention it;
- *compositional*: it needs two or more mechanisms working together.

Not every CogLean theorem is a psychological explanation. Some are properties of algorithms, needed library facts that explain nothing on their own; some are relations between laws; some, such as confident errors and solver independence, are compositional. Making these distinctions visible is part of the method: Lean shows every premise, and the judgment of what the premises contain is written down, in the open.

8 Checking the formalization

Lean guarantees that the theorems follow from the definitions. It does not guarantee that the definitions say what the prose model says. We tested that link with an independent implementation that could not simply copy the first [cf. N-version programming, 14].

Method. The model was written as a prose specification, with no Lean names, theorems or predictions (`MultiDigit/sim/SPEC.md`). Points the prose leaves open were listed for the implementer to settle and document. Before any simulator existed, a list of checks was committed (`PREREG.md`). Each check named a theorem, the observable statistic, the hypothesis as read from the simulator's logs, and a tolerance: zero violations for exact relations, and frequencies within 4σ for probabilities. A check whose hypothesis never arose was to be reported as vacuous, not as passed. A fresh AI agent, given only the specification and working in an empty directory, wrote the simulator. An audit of its tool calls showed that it read only files it had written. Lean exported exact trial distributions for four instances, and a harness ran the checks on 30 random fixed instances and 20 random learning instances.

Agreement. Every check whose hypothesis arose passed, with no failures. 243 outcome frequencies matched Lean's exact distributions. The predictions about routes, confident errors, carrying, column errors, zeros and thresholds, and learning (non-interference, discrimination, then column work) passed on hundreds to hundreds of thousands of tests each. The simulation predates the problem solver. Because the solver is proved equal to the column procedure the simulation implements (`Solver.bridge`), its results carry over.

What the simulation found that the proofs did not. Three findings remain open in the current library.

1. **Hypotheses unrealized in the tested conditions.** Five predictions (near misses within 200 or 20, round problems, and column facts) were vacuous in the simulation: their hypotheses never held in any of the nets EPAM learned across the simulation's random instances. Each needs a path that tests *every* feature of a digit or column, and EPAM grows one test per confusion, so paths in these histories stayed short. This does not show that the hypotheses are unreachable. No impossibility theorem exists, and a long enough history of confusions on one digit could plausibly produce such paths. It shows that the theorems say little about the nets these learning histories produced. Proof alone did not reveal this; counting how often each hypothesis arose did. The distinction is between logical correctness and the model states that learning actually produces.

2. **A gap at zero strength.** With associations of total strength 0, sampling divides by zero, and a trial has total probability 0. The theorems avoid the case by assuming positive totals. The simulator, reading the prose, treats it as exhausted retrieval, which is the natural reading.

3. **An unstated assumption decides a predicted learning sequence.** The combined model suggests a sequence: a confident error, then, after discrimination, slow column work, then fluent correct retrieval. Whether it occurs depends on the initial associations of a newly discriminated problem, which the prose left open. With none, 40 of 48 discriminated problems followed the sequence. With only the correct answer, the column phase was skipped. With a copy of the neighbour's associations, the error was stated, strengthened and stated again:

7 of 44 problems were trapped, and 35 were still mixed at the end. The blind implementer flagged this dependence from the specification alone.

The simulation does not prove the formalization faithful to the intended theory. It is independent evidence that the formal model behaves as its prose describes, and it found where the formal model says too little. Three standards are involved: logical correctness (Lean), behavior in the states learning produces (simulation), and empirical adequacy (data about people), which neither addresses.

9 Discussion

9.1 Relation to previous formal work

CogLean is not the first formalization or mechanical verification of a cognitive theory. Formal specification of cognitive architectures goes back at least to a specification of Soar in Z [15]. It covered the architecture completely but at a low level, and it could not be executed. Cooper, Fox, Farringdon and Shallice argued that better theories would come from languages designed for specifying cognitive models, and gave one, Sceptic, an executable specification language in which they implemented several models, including Soar [16]. Stewart and West rebuilt ACT-R in order to separate its core commitments from implementation detail and to explore alternative components [17]. That anticipates CogLean's interest in interchangeable mechanisms, in an executable setting. Ragni and colleagues proposed a formal framework into which architectures such as ACT-R and Soar can be embedded for comparison [18].

For ACT-R in particular, Gall and Frühwirth gave an abstract operational semantics, a translation to Constraint Handling Rules proved sound and complete with respect to it, and a decidable confluence test [19, 20]. Langenfeld, Westphal and Podelski translated ACT-R models to timed automata for exhaustive defect analysis [21], and proved their timed-automaton semantics correct at the level of architectural components [22]. Macklem and Mili proposed specification languages and algorithms for checking cognitive models against requirements [23]. In the context of the Common Model of Cognition [24], Ganeriwala and colleagues have proposed enabling formal verification of cognitive models [25]. They have also compiled Soar models to discrete-time Markov chains for probabilistic model checking in PRISM [26, 27], which addresses stochastic behavior, as CogLean's exact distributions do. Outside architectures, verbal theories have been formalized as logic programs [28] and as communicating automata [29]. Fisher and colleagues derived critical tests of ACT-R from its core architectural assumptions, as theorems over a space of model classes [30], an aim close to CogLean's separation of what follows from a theory's core.

Model verification and theoretical deduction. Three approaches to establishing what a cognitive model does should be kept apart.

- **Simulation** samples executions to estimate behavioral outcomes.
- **Probabilistic model checking**, as in PRISM, analyzes a specified state-transition model exhaustively, often by numerical methods, to establish properties or compute probabilities over its whole state space. This is formal verification, not simulation. It can also treat parameters symbolically, giving a property or probability as a function of them [31].
- **Deductive theorem proving**, as in CogLean, establishes results for whole classes of models that satisfy stated theoretical assumptions. They are independent of particular parameter settings and of any finite enumeration of states.

The important distinction is therefore not between simulation and formal verification, but between *model verification* and *theoretical deduction*. Model checking primarily verifies properties of a particular formalized system, or of a parameterized family of them. CogLean's principal objects are the theories themselves: reusable representations, mechanisms, assumptions and theorems. Its aim is to establish general consequences of those theories, and to derive further results by composing independently formalized mechanisms. The approaches are complementary, not competitors. A model checker can verify a large, detailed model that would be laborious to reason about deductively. A theorem can state what every such model must do, and why.

CogLean does not introduce formal methods to cognitive science. What is distinctive, relative to this work, is a combination of emphases, few of them entirely new on their own:

- **Theorems over classes of models**, not properties of one model or one parameterized family. CogLean's theorems quantify over every encoding, net, strength set and backup, and, through laws, over every memory.
- **Phenomena and laws stated theory-neutrally**, so that rival mechanisms can be tested against the same statement.
- **Reusable proofs across independently developed theories.** The previous work formalizes one architecture or model at a time. CogLean's central example composes theorems proved separately for EPAM, DOA and a problem solver into results about their combination.
- **Behavioral consequences derived from composition**, with their premises sorted into postulates, conditions and settings.

Whether any single one of these is novel is a matter for a more complete review. What we claim to be new is their combination: a shared, cumulative library of cognitive theory that supports reusable proofs, composition across theories, and deductive discovery.

9.2 Relationship to cognitive architectures

CogLean is not proposed as an alternative to cognitive architectures such as Soar and ACT-R, or to efforts such as the Common Model of Cognition [24]. It offers a possible formal foundation for expressing, composing and reasoning about their constituent mechanisms and architectural commitments. The present example composes mechanisms at a relatively local level, for one task, without committing to a general cognitive architecture. A longer-term objective is to formalize architectural constraints themselves, so that their implications can be proved and alternative architectures compared within a shared framework.

9.3 What the approach bought

The examples above show four kinds of gain, of different weight. Some results are *library facts*, such as the bridge and the well-formedness of the composed distributions: unglamorous, but they are what lets theorems transfer. Some are *compositional deductions*, true at every setting: whole-problem errors do not depend on the solver. Some are *conditional*, and the conditions are where the science is: confident errors need confusion and sharp associations, and the proofs say exactly which premise a failed test would implicate. And some are *corrections*: the DOA chapter's claim about peaking, the theorem hypotheses that learning did not realize, and the unstated assumption behind the predicted learning sequence. None of these is yet an empirical result about people.

Proof and simulation were complementary here, in a way the slogan "simulation explores, proof settles" understates. The simulation checked what the proofs are *about*: whether their hypotheses arise, and whether the definitions say what the prose says.

9.4 Limitations

- The associative strengths in the worked example are chosen, not derived from experience. Siegler and Shrager's priors came from a counting process. Grounding these in learning histories is open.
- The strongest results depend on sharp associations and on particular recognition confusions. Confusions are derived for learned nets; sharpness is not.
- Single-digit DOA assumes that a representation with the right number of objects is available for elaboration and counting. As the column process of a multi-digit adder, its fallback to fingers and counting is implausible: by then single-digit facts are retrieved. The fallback should be a parameter of the column process; the general theorems do not depend on it.
- ACT-R's memory is represented by its retrieval rule, without retrieval as use: memory changes when an item is studied, not when it is retrieved during a trial. Every ACT-R value here, including Table 1, §5.5 and the counterexample to the pattern, holds for that version. Full ACT-R, in which each retrieval strengthens what it retrieves, would give different values, and possibly different theorems.
- Carrying is error-free in the full procedure, and columns are added independently.
- The framework does not establish psychological plausibility. That remains a matter of evidence and judgment.
- Scalability is a serious question. CogLean uses exact rational distributions over finite outcome spaces. Continuous representations, large state spaces and neural models will need different techniques, and exact evaluation already required a merged evaluator here.
- Three findings of the simulation remain open: hypotheses unrealized in the tested learning histories, the zero-strength gap, and the initial associations of a newly discriminated problem.
- CogLib's distributions are rational-weighted lists; the type does not enforce nonnegative weights. Most library theorems are algebraic identities that hold regardless, but reading them as probabilities needs nonnegativity. It is proved here for the integrated model, given nonnegative strengths and rates, but not yet across the whole library.

9.5 Future work

Natural next steps:

- derive associative distributions from learning histories, so that sharpness becomes a theorem;
- add feedback-driven learning across whole trials, so that the predicted learning sequence can be proved;
- restate the vacuous predictions in terms of the features a net actually tests, or prove when learning can realize their hypotheses;

- generalize the retriever interface, so that more memories can be swapped in, and so that retrieval can change memory (ACT-R's retrieval as use), with the memory's state carried through the solver;
- compose theories in other domains.

The library already contains early work on the last of these: conceptual combination, semantic memory and view application, on a shared semantic-network representation.

9.6 Conclusion

The question was whether theories represented in a common formal library can be composed to yield meaningful consequences that would otherwise require building a new model from scratch. In one case the answer is yes. EPAM's recognition, DOA's retrieval and strategy choice, and a decision-cycle problem solver, each formalized on its own, compose into a model of three-digit addition with no answer supplied, and its consequences are proved largely by reusing the components' theorems. The larger ambition is a cumulative theoretical science, in which cognitive constructs and their proved properties are reused across models, tasks and research traditions, as mathematical results are reused in Mathlib. One example does not establish that. It shows that the approach can produce results of the right kind.

Code. All definitions and theorems are in the CogLean repository [7], `https://github.com/jeffshrager/CogLean`. Every theorem is fully proved, with no `sorry`, and depends only on Lean's standard axioms.

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A Index of results

Claim	Lean name (file)
Discrimination learning preserves prior recognition	`learn_preserves` (`CogLib/Process/DiscriminationLearning`)
A confusion agrees on every tested feature	`confusion_agrees_on_path` (`CogLib/Memory/DiscriminationNet`)
Learned nets recognize each image as itself	`learnAll_clean` (`CogLib/Process/DiscriminationLearning`)
DOA retrieval use, error share, backup independence	`SC.retrieval_prob`, `retrieval_error_share`, `retrieval_independent_of_backup`
Counting errors independent of associations	`SC.counting_error_share`
Peaking never raises retrieval errors; but can raise overt strategy use	`SC.peaking_lowers_retrieval_errors`; `SC.peaking_can_raise_overt_use`
The phenomena: confusion-error pattern; confident errors with a bound	`ConfusionErrorPattern`, `ConfidentErrors` (`CogLib/Phenomena/ConfusionErrors`)
Confident errors, any task; with the rate as bound	`RR.confident_errors`, `RR.confident_errors_rate` (`CogLib/Process/RecognizeRetrieve`)
Confident errors from laws of memory	`RR.confident_errors_of_laws`
Solver equals the column procedure	`Solver.bridge` (`MultiDigit/Solver`)
A column error is a DOA error in place	`MD.column_error_is_doa_error` (`MultiDigit/Columns`)
Integrated: solver is the column backup; mass 1; nonnegative weights	`solverBackup_eq_columns`, `mass_solverBackup`, `nonneg_solverBackup`, `nonneg_itrial` (`MultiDigit/Integrated`)
Integrated: rate of solver use, and its bound	`integrated_backup_rate`, `integrated_backup_le`
Integrated: whole-problem errors independent of solver	`whole_independent_of_solver`
Integrated: column route factored	`columnsSuch_prob`
One memory: column fact is DOA under self-recognition; reduction to the integrated model	`memFact_self`, `strial_reduces`, `strial_net_reduces` (`MultiDigit/SharedMemory`)
One memory: confident errors on digit pairs; values at the setting	`column_confident_errors`; `setting_epam_values`, `setting_act_pairs`, `setting_act_values`
Integrated: confident errors; with a bound; two sources	`integrated_confident_errors`, `integrated_confident_errors_le`, `errors_two_sources`
Integrated: solver error is a DOA error (other columns exact); exact facts sufficient, not necessary	`solver_error_is_doa_error`, `solver_exact_of_exact`, `compensating_errors`
Integrated: values at the setting (EPAM; ACT-R)	`setting_routes`, `setting_solver_x`, `setting_overall_x`, `setting_overall_q1`, `setting_practiced_errors`, `setting_valid`, `setting_confident_errors_bound`; `setting_actr_x`, `setting_actr_not_pattern`
ACT-R: graded confusion; interference	`act_values`, `act_interference` (`MultiDigit/ExampleACTR`)
EPAM: practice never disturbed	`MD.learning_preserves_trial`
Memory laws: EPAM has, ACT-R never has	`Matrix.Discrimination.*`, `Matrix.Activation.never_*` (`CogMatrix`)
Merged evaluation is the same distribution	`SC.same_trialMerged`, `same_fastBackup`